

Allison Otero

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Professional Summary

Biomedical Engineering student with experience in AI/ML research, robotic systems, embedded sensing, biomedical instrumentation, and physiological signal analysis. Interested in aerospace, healthcare technology, human factors, systems engineering, robotics, and AI-embedded systems.

Education

Bachelor of Science: Biomedical Engineering, Minor in Biology Expected May 2027
Florida Institute of Technology – Melbourne, FL

Certificate: Supervised Learning: Regression and Classification July 2025
Stanford University – Stanford, CA

Associate of Arts May 2023
Eastern Florida State College – Melbourne, FL

Skills

Programming & Data Analysis: Python, MATLAB, Microsoft Office (Excel, Word, PowerPoint), statistical analysis, predictive modeling, data visualization.

AI/ML: Supervised machine learning, neural network development, feature extraction, model testing, TinyML/embedded AI.

Biomedical & Signal Processing: sEMG, MyoWare sensors, physiological signals, biomedical instrumentation, signal processing, sensor-based data collection.

Embedded Systems & Hardware: Arduino Uno, Arduino Nano 33 BLE Sense, Raspberry Pi, sensor interfacing, hardware-software integration, real-time systems.

Engineering & Prototyping: CAD, manufacturing, machine shop, waterjet, CO2 and fiber laser cutter.

Professional Skills: Technical communication, teamwork, organization, public speaking, leadership, project coordination.

Experience

AI/ML Research Assistant | Florida Institute of Technology May. 2026 – Current

- Conduct AI/ML research focused on robotic systems, embedded intelligence, and neural network development for real-time edge applications.
- Build and evaluate machine learning models using Python, sensor-based datasets, signal-processing, and feature extraction techniques.
- Support embedded AI deployment using Arduino, Raspberry Pi, and TinyML for applications in robotics, health monitoring, and smart sensing.
- Assist with hardware-software integration, sensor interfacing, data collection, model testing, and technical troubleshooting.

Robotics Member | Florida Institute of Technology Aug. 2025 – Current

- Assisted in integrating mechanical, electrical, and software subsystems to ensure reliable robot performance.
- Gained hands-on experience in robotics system integration, rapid prototyping, and engineering problem-solving to meet the deadlines and deliverables required for NASA Competition.

Data Analysis Intern | Defense Informatics LLC Jun. 2025–Aug. 2025

- Conducted comprehensive statistical analysis, uncovering hidden trends and patterns within the data.
- Implemented machine learning algorithms for predictive modeling, resulting in better decision-making capabilities.
- Utilized data visualization techniques to present and explain complex data sets.

Hospital Volunteer | Viera Hospital Jun. 2025–Nov. 2025

- Prepared surgical equipment for clinical use by inspecting instruments, maintaining sterile handling practices, and assisting with proper storage and distribution.
- Gained hands-on healthcare experience while developing attention to detail, procedural accuracy, teamwork, and an understanding of hospital safety protocols.

Student Ambassador | Florida Institute of Technology Dec. 2023 – Current

- Developed essential leadership skills through public speaking experience and regular presentations to diverse audiences about the benefits of attending the university.